Neural networks trained on compact group composition learn the global form of the group

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Abstract

Small neural networks trained to identify products in a compact Lie group, observed only through a generic scrambled encoding, build the irreducible representations of the group's global form: across ten group forms drawn from $\mathrm{SU}(2)$, $\mathrm{SU}(3)$, $\mathrm{Spin}(4)$, and $\mathrm{Spin}(5)$ and their central quotients, the learned code concentrates on a small faithful carrier determined by the quotient, with half-integer spins present exactly when the data distinguish the double cover, deterministically across seeds. The code is the full matrix-coefficient block of its carrier and supports commuting linear left and right translation actions that smaller faithful frames provably lack. In a matched comparison on $\mathrm{SO}(3)$, the block attains a larger margin than a faithful frame at 17% lower parameter norm, so training selects it as the cheaper complete mechanism. Under regression onto heat-kernel targets, the same architecture fills isotypic sectors in order of Casimir eigenvalue, with a truncation frontier set by the weight decay rather than by capacity.

1 Introduction

A compact Lie group carries a canonical harmonic analysis: by the Peter–Weyl theorem, the matrix coefficients of its irreducible unitary representations span a dense subspace of L^2 [Bröcker and tom Dieck, 1985, Ch. III]. For *finite* groups, mechanistic interpretability has established that small networks trained on the multiplication table represent elements through matrix coefficients of a few irreducible representations [Chughtai et al., 2023, Sec. 4], that competing descriptions of the same models exist [Stander et al., 2024, Sec. 5], and that per-argument equivariance reconciles them Wu et al. [2025]. All of this work presupposes a finite table and one learned embedding per element, the setting of the grokking literature Power et al. [2022], Nanda et al. [2023], Gromov [2023], Zhong et al. [2023], and the associated theory proves irreducible features emerge for finite groups only Morwani et al. [2024], Marchetti et al. [2024], while equivariant architectures place the harmonics into the network by construction Cohen and Welling [2016], Geiger and Smidt [2022]. Whether this representation theory emerges in a generic network when the group is a continuum has remained open.

In this paper, we train encoder-factorized scorers to identify products in compact groups presented only through frozen generic observation maps, and we analyze the learned representations. Three results follow.

The learned carrier identifies the global form of the group. For each central subgroup $Z \subseteq Z(G)$, averaging the observation map over Z realizes the same task on the quotient G/Z. Across ten group forms (Table 1), the encoder concentrates 93–98$\%$ of its variance on a small faithful carrier of the task group, identically in every seed, with forbidden sectors at the measurement floor, and ablating the carrier destroys exactly the discriminations its central character supports (Section 3).

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The learned code is the full two-sided coefficient block. The active subspace has exactly the real dimension $(\dim_{\mathbb{R}} W)^2 / \dim_{\mathbb{R}} \mathrm{End}_G(W)$ of the full coefficient block of its carrier W, and fitted left and right translation operators fit at held-out $R^2 \geq 0.98$ and commute, which characterizes the full block within its sector (Lemma 2). A faithful frame passes the left test and fails linear right closure, and in matched training the block attains a larger margin at 17% lower parameter norm (Section 4): optimization selects the block not as the smallest separating code, which it is not, but as the cheapest complete mechanism.

Regression targets reveal a Casimir-ordered spectral bias with a regularization-set frontier. Trained to regress heat-kernel values $K_t(ab\,c^{-1})$, the same architecture builds multi-sector codes whose score-level spectral profile recovers the required sectors at their required weights up to a sharp cutoff, the cutoff advances sector by sector in order of Casimir eigenvalue during training, and on a band-limited target the in-support sector that weight decay 1.0 prunes is retained at weight decay 0.1 (Section 5).

Our methods are standard. What is new is the object, a continuum group with no embedding table, the quotient construction that turns the global form into an experimental variable, and the two-sided translation test of Section 4. The rest of this paper is organized as follows. In Section 2, we fix the task and the spectroscopy. In Sections 3 to 5, we present the carrier, mechanism, and spectral results. In Section 6, we state the open problems. Proofs and experimental details appear in the appendices.

2 Task, observations, and spectroscopy

Throughout, G denotes a compact Lie group with Haar probability measure, $Z \subseteq Z(G)$ a central subgroup, W an irreducible real representation of G with commutant $\mathrm{End}_G(W) \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, $z(g) \in \mathbb{R}^{128}$ the trained argument-encoder activations, and χ_ρ the character of an irreducible unitary representation ρ.

Task. Given encodings of $a, b \in G$ and of $K = 64$ candidates, the model must identify the product. The positive candidate is $(ab)\exp(\xi)$ with ξ a centered Gaussian of scale $\sigma = 0.2$ on the Lie algebra, the $K - 1$ negatives are Haar samples excluded from a ball around the positive class, and the loss is cross-entropy over the K scores. Training pairs come from a pool of 2048 Haar elements, and we report ranking accuracy on pairs of fresh elements never seen in training.

Observations. Each element is presented through a frozen map Ψ rather than a token, for $\mathrm{SU}(2)$ the map $\Psi(q)_i = \exp(\alpha\langle q, h_i\rangle)$ with 32 fixed Haar landmarks h_i and $\alpha = 3$, standardized under Haar statistics. The quotient condition for G/Z replaces Ψ by its average over the Z-orbit, which is exactly Z-invariant, so the observations determine elements of G/Z and nothing finer, with all else shared. Every result below reproduces under two further observation maps (Appendix C), so the selected carrier is a property of the task and the group.

Model. An encoder-factorized scorer with about 0.3M parameters: a shared two-layer GELU encoder produces $z(a), z(b)$, an interaction network produces $h(a, b) \in \mathbb{R}^{128}$, a candidate encoder produces $z_c(c)$, and the score is $\langle h(a, b), z_c(c)\rangle$, trained by AdamW with weight decay 1.0 unless stated. Every reported quantity is a mean over five seeds with its standard deviation.

Spectroscopy. For a fixed probe set of Haar samples, we compute the fraction of variance of the centered activations z explained by the sampled, jointly orthonormalized spans of matrix coefficients, sector by sector in Casimir order, with exact coefficients verified in a unit-test suite (Appendix C) and a sampling floor near 0.2%. Activations are real, so a complex irreducible and its conjugate span one real sector of dimension $(\dim_{\mathbb{R}} W)^2 / \dim_{\mathbb{R}} \mathrm{End}_G(W)$ (Lemma 2).

3 The learned carrier identifies the global form

Across the ten group forms of Table 1, the encoder concentrates on a small faithful carrier of the task group, deterministically across seeds. The half-integer sector of $\mathrm{SU}(2)$ appears if and only if the observations distinguish $\pm g$, with $\cos(z(g), z(-g)) = -0.95 \pm 0.02$ against $+1.000$ in the quotient condition, so the learned code detects the double cover, a property invisible to the Lie algebra. When the center is not cyclic, no faithful irreducible representation exists, and the models build the forced direct sum. Where a smaller reducible faithful sum competes against a larger faithful irreducible representation, the sum wins, with the alternative under 1% of variance in every seed.

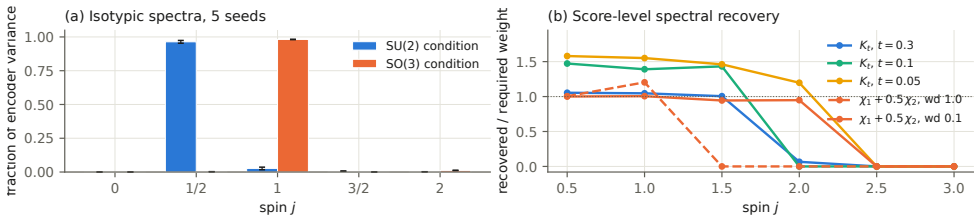


Figure 1: (a) Isotypic spectra of the trained encoder for the $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ conditions (five seeds). (b) Score-level spectral recovery for heat-kernel and band-limited targets. Values near 1 indicate a sector carried at its required weight, and the dashed curve shows the in-support spin-2 sector pruned at weight decay 1.0.

Table 1: Learned carriers across ten group forms, with the fraction of encoder variance on the carrier (mean over five seeds) and the rejected faithful alternative where one exists. $\mathrm{Spin}(4)$ sectors are pairs (j_L, j_R).

task group	learned carrier	variance	rejected alternative
$\mathrm{SU}(2)$	spin $\frac{1}{2}$	0.96	
$\mathrm{SO}(3)$	spin 1	0.98	
$\mathrm{SU}(3)$	$\mathbf{3} \oplus \bar{\mathbf{3}}$	0.97	
$\mathrm{PSU}(3)$	adjoint $\mathbf{8}$	0.93	
$\mathrm{Spin}(4)$	$(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$	0.98	(no faithful irrep)
$\mathrm{SO}(4)$	$(\frac{1}{2}, \frac{1}{2})$	0.97	
$\mathrm{SO}(3)\times\mathrm{SU}(2)$	$(1, 0) \oplus (0, \frac{1}{2})$, 5 dims	0.98	$(1, \frac{1}{2})$, 6 dims: 0.008
$\mathrm{PSO}(4)$	$(1, 0) \oplus (0, 1)$, 6 dims	0.98	$(1, 1)$, 9 dims: 0.009
$\mathrm{Spin}(5)$	spinor $\mathbf{4}$	0.97	
$\mathrm{SO}(5)$	vector $\mathbf{5}$ + adjoint $\mathbf{10}$	$0.69 + 0.22$	(see Section 6)

The selection is causal. Subtracting from z the component linearly predictable from the carrier's matrix coefficients collapses ranking to chance (0.02 against a chance rate of 0.016) and center discrimination to its tie value, while the same ablation applied to any other sector, or to a random subspace of equal rank, leaves both intact. The pattern forms a double dissociation between each group and its quotient: for $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$, ablating spin $\frac{1}{2}$ is fatal for the former and inert for the latter, and ablating spin 1 transposes the pattern, with seed standard deviations below 0.005 on every cell. The $\mathrm{SU}(3)$ pair repeats this under the order-three center, where the orbit-averaging intervention is an exact identity on the quotient models and fatal on the faithful ones. For $\mathrm{Spin}(4)$, whose center supports three distinct discriminations, each learned block is necessary for exactly the center flips its central character detects and for no others. The trained scorer itself is a character function: a single character, $\chi_{1/2}$ for $\mathrm{SU}(2)$ and χ_1 for $\mathrm{SO}(3)$, explains 0.97 ± 0.01 and 0.88 ± 0.01 of the score variance on held-out triples.

The following bound, proved in Appendix A, explains why ranking never forces a larger carrier.

Lemma 1. *Let V be a faithful finite-dimensional unitary representation of a compact group H with character χ_V. Then $\mathrm{Re}\,\chi_V(g) \le \chi_V(e)$ for all $g \in H$, with equality only at $g = e$. Moreover, for X in the Lie algebra of H, $\mathrm{Re}\,\chi_V(\exp X) = \chi_V(e) - \frac{1}{2}\|dV(X)\|_F^2 + O(\|X\|^4)$, and $X \mapsto \|dV(X)\|_F^2$ is positive definite.*

By Lemma 1, a monotone function of $\mathrm{Re}\,\chi_V(ab\,c^{-1})$ ranks the product above every excluded negative for any faithful V, so optimization selects the carrier on cost. Conversely, a nontrivial kernel forces positive ranking error under explicit sampling assumptions (Lemma 3), and the ablations above realize that necessity block by block.

4 The mechanism is the full coefficient block

The learned code is not merely supported on the carrier's isotypic sector. In every seed, the active subspace of z has exactly the real dimension of the full block, 4 for the quaternionic spinor and 9 for the real-type vector, as the following lemma prescribes (proof in Appendix A).

Lemma 2. *Let $\rho\colon G \to U(V_\rho)$ be an irreducible unitary representation of a compact group G, and let $M_\rho \subseteq L^2(G)$ be the span of its matrix coefficients $g \mapsto \langle u, \rho(g)v\rangle$. Then $M_\rho \cong V_\rho^* \boxtimes V_\rho$ as a*

representation of $G \times G$ acting by left and right translation, this representation is irreducible, and every nonzero subspace of M_ρ invariant under both translation actions equals M_ρ. For an irreducible real representation W with commutant $D = \mathrm{End}_G(W)$, the corresponding real coefficient block has dimension $(\dim_{\mathbb{R}} W)^2 / \dim_{\mathbb{R}} D$ and is the minimal nonzero two-sided translation-stable real subspace of its sector.

Two-sided structure. Lemma 2 converts full-block identity into a falsifiable property: a full block supports commuting linear left and right translation actions, a proper subspace of the sector cannot. We fit affine operators M_k^L, M_k^R with $z(kg) \approx M_k^L z(g)$, $z(gk) \approx M_k^R z(g)$, evaluated held out. On the trained $\mathrm{SO}(3)$ code, both actions fit at $R^2 = 0.989 \pm 0.001$ with relative commutator defect 0.012 ± 0.001, matching the exact block. On the explicit 6-dimensional frame $g \mapsto (ge_1, ge_2)$, the left action is exactly linear and the right action fails at $R^2 = 0.638$, since the missing column $ge_3 = (ge_1) \times (ge_2)$ is uncorrelated with the retained ones, and a random 6-dimensional subspace of the true block fails both actions at $R^2 = 0.67$. The fitted operators match the representation itself: the trace of M_k^L equals $\dim(W)\, \chi_W(k)$ with correlation 1.000 across group elements.

Causal sufficiency. Replacing the internal product state $h(a, b)$ by its ridge reconstruction from the eight realified spinor coefficients of a counterfactual product $a'b'$ redirects the $\mathrm{SU}(2)$ model to select $a'b'$ with probability 0.996–0.999 in every seed, while reconstruction from the wrong sector redirects at chance. The output h concentrates $89.9 \pm 1.4\%$ of its variance on the diagonal sector of its carrier, with $98.2 \pm 0.3\%$ of that energy in the subspace spanned by the matrix product $W(a)W(b)$: the interaction implements the representation's own composition law.

Cost. Whether two-sided closure is why the block wins is a question about training, so we trained matched models on $\mathrm{SO}(3)$ whose observations are the exact frame or the exact full block, all else equal. At statistically equal accuracy (0.891 ± 0.002 against 0.889 ± 0.003), the block model attains a larger margin (7.29 ± 0.16 against 6.49 ± 0.05) at 17% lower parameter norm, with every module cheaper: the frame model reconstructs the missing column nonlinearly, at a cost exceeding the three saved coordinates. Regularized training selects the full block as the cheapest complete mechanism, storage and composition counted together.

5 Casimir-ordered spectral bias under regression

Ranking requires only one faithful character, so it cannot probe how the architecture allocates capacity across sectors. Regression onto the $\mathrm{SU}(2)$ heat kernel $K_t(g) = \sum_j (2j+1) e^{-t\, j(j+1)} \chi_j(g)$, evaluated at $ab\,c^{-1}$, requires every sector the envelope keeps above the noise. Figure 1(b) reports the fraction of score variance per sector, sequentially orthogonalized in Casimir order, divided by the same fraction for the exact target, so 1 means a sector at its required weight.

Three regularities emerge. Each trained score is a truncated character expansion, with recovered sectors near ratio 1 and sectors beyond a sharp frontier at 0, the frontier at higher Casimir for smaller t. The frontier advances in Casimir order during training: at $t = 0.05$, the spin-2 recovery climbs from 0.12 to 1.20 over 70k steps while spin $\frac{5}{2}$ stays at 0. The frontier is set by regularization, not capacity: on the target $\chi_1 + \frac{1}{2}\chi_2$, weight decay 1.0 prunes the spin-2 component and the abandoned 12.6% of target variance equals the final test error, while weight decay 0.1 recovers the target exactly. A sector survives when its variance contribution exceeds its norm cost, so the spectral cutoff is a cost boundary of the same kind that selects the carrier. The target's central character governs parity: on this central-even target, the model builds an integer-spin code even on the faithful encoding.

6 Limitations and open problems

The minimal-carrier description fails as stated in one of the ten group forms: on $\mathrm{SO}(5)$, the learned carrier is a mixture, vector $\mathbf{5}$ at 0.68–0.70 of variance plus adjoint $\mathbf{10}$ at 0.21–0.23, stable across five seeds, doubled training, more candidates, and reduced weight decay. The adjoint fraction falls only when generalization improves and no condition solves the task, so asymptotic purification and a genuinely mixed minimum-cost mechanism both remain open. In Appendix B, we state the selection principle these results suggest as a precise conjecture, and we leave the conditioning of quotient observations as future work.

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A Proofs

Proof of Lemma 1. Each eigenvalue of the unitary matrix $V(g)$ lies on the unit circle, so $\operatorname{Re}\operatorname{tr} V(g) \le \dim V$, with equality exactly when every eigenvalue equals 1, that is, when $V(g)$ is the identity. By faithfulness, this occurs only at $g = e$. For the expansion, write $A = dV(X)$, an anti-Hermitian matrix with eigenvalues $i\lambda_1, \dots, i\lambda_{\dim V}$. Then $\operatorname{Re}\operatorname{tr}\exp(tA) = \sum_k \cos(t\lambda_k) = \dim V - \frac{t^2}{2}\sum_k \lambda_k^2 + O(t^4)$, and $\sum_k \lambda_k^2 = \|A\|_F^2$. The trace of A is purely imaginary and contributes nothing to the real part. If $\|dV(X)\|_F = 0$, then the one-parameter subgroup $\exp(tX)$ lies in the kernel of V, which is trivial, so $X = 0$, proving positive definiteness. □

Lemma 3. *Let H be a compact group. Assume the score factors as $s(a, b, c) = S(\varphi(a), \varphi(b), \varphi(c))$ where φ factors through the matrix coefficients of a representation V of H with nontrivial closed kernel N, the function $c \mapsto s(a, b, c)$ is continuous, negatives are sampled from a distribution assigning positive probability to every open subset of the admissible region, and the exclusion radius is smaller than $\min_{n \in N,\, n \neq e} d(e, n)$. Then the population ranking loss is bounded away from its optimum.*

Proof. Fix $n \in N$ with $n \neq e$. By the radius hypothesis, $ab\,n$ lies in the admissible negative region. Since φ is constant on N-cosets, $s(a, b, ab\,n) = s(a, b, ab)$, which is at least the score of the noisy positive up to the continuity modulus of s over the positive displacement. By continuity, there is a ball around $ab\,n$ on which the score stays within any prescribed gap of this value, and by the sampling hypothesis that ball receives positive probability, so with positive probability a sampled negative ties or exceeds the positive. □

The hypotheses of Lemma 3 are necessary: with a finite kernel and a continuous candidate distribution, the exact ambiguous points have measure zero, and without continuity of the score no positive lower bound follows from non-separation alone.

Proof of Lemma 2. The map $V_\rho^* \otimes V_\rho \to L^2(G)$ sending $u^* \otimes v$ to $g \mapsto \langle u, \rho(g)v\rangle$ intertwines the $G \times G$-action by left and right translation with the outer tensor product action and is injective by linear independence of matrix coefficients [Bröcker and tom Dieck, 1985, Ch. III, Thm. 1.5]. The outer tensor product of irreducible representations is an irreducible representation of the product group, so M_ρ has no proper nonzero two-sided invariant subspace, and $\dim_{\mathbb{C}} M_\rho = (\dim V_\rho)^2$. For the real form, the real Schur lemma gives $\mathrm{End}_G(W) \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, and the real matrix-coefficient span of W is identified with $W^* \otimes_D W$, of real dimension $(\dim_{\mathbb{R}} W)^2 / \dim_{\mathbb{R}} D$. Minimality follows from irreducibility of the two-sided action on this space, by the same outer-tensor argument applied over D. □

B The selection conjecture

Conjecture 1. *In the factorized architecture with weight decay, among parameter settings achieving population ranking loss within ε of the optimum, the minimum-norm setting realizes, as $\varepsilon \to 0$, the full real coefficient block of a faithful carrier minimizing the total regularized cost of features, composition, and scoring, with score a monotone function of* $\mathrm{Re}\,\chi_V(ab\,c^{-1})$.

The missing ingredients are a norm lower bound tying any ε-optimal score to the effective dimension of the functions it computes on G, and the exclusion of solutions outside the character algebra, the compact-group analogue of the alternative circuits documented for finite groups Stander et al. [2024], Zhong et al. [2023]. The descent rule, the lemmas of Appendix A, and the measured cost gap between frames and blocks fix every term such a proof must reproduce.

C Experimental details

Groups and encodings. Elements of $\mathrm{SU}(2)$ are unit quaternions, of $\mathrm{SU}(3)$ Haar-random special unitary matrices, of $\mathrm{Spin}(4) = \mathrm{SU}(2) \times \mathrm{SU}(2)$ pairs of unit quaternions, and of $\mathrm{Spin}(5) = \mathrm{Sp}(2)$ quaternionic 2×2 unitaries sampled by quaternionic Gram–Schmidt. Quotient conditions average the observation map over the central subgroup being quotiented: the antipodal average for $\mathrm{SO}(3)$ and $\mathrm{SO}(5)$, the order-three orbit average for $\mathrm{PSU}(3)$, and the three sign-pair averages for the quotients of $\mathrm{Spin}(4)$. Each average is exactly invariant under the quotiented subgroup and distinguishes all other center elements, which we verify numerically at construction.

Training. Batch size 4096 (2048 for the larger groups), learning rate 10^{-3}, weight decay 1.0, 100k–200k steps, training fraction 0.01 of the pair table unless stated. The ranking results of Table 1 use five seeds per condition. Heat-kernel and band-limited regressions replace the cross-entropy by mean squared error on standardized targets, with candidates drawn half near the product at uniform geodesic radius and half from Haar measure. Under the $\mathrm{SU}(2)$ sampling, the Bayes-optimal ranking rule attains accuracy 0.951.

Analyses. Isotypic fractions use jointly orthonormalized sampled coefficient spans processed in Casimir order, which bounds cross-sector leakage at the sampling floor. Function-space ablations subtract the component of z linearly predictable from a sector's exact matrix coefficients, fitted on an independent probe set, and are well defined even for sectors at the measurement floor, where subspace estimates are noise-dominated. Ridge probes select the regularization strength on a validation split and report on a disjoint test split. Translation operators are fitted by ridge regression in a PCA frame retaining 99.9% of activation variance, and homomorphism and commutator defects are

evaluated on held-out elements with operators fitted independently per group element. The score-level spectral profiles regress both the learned score and the exact target on the same sequentially orthogonalized character design, so standardization and collinearity cancel in the reported ratios.

Verification. The representation-theoretic primitives pass a unit-test suite covering the homomorphism property, unitarity, characters, central characters, Schur orthogonality, planted-sector recovery, and exact invariance of the quotient encodings. The spectral-recovery analysis reproduces the known support and coefficients of the band-limited target at weight decay 0.1, which serves as the calibration of the instrument. The orbit-average intervention is an exact identity on quotient models by construction and measures as one, which confirms the intervention machinery. Software: PyTorch 2.11, NumPy, single T4 GPU. A tracked results bundle records, for every run, the configuration, seed, checkpoint hash, final metrics, and analysis outputs, and every table in this paper regenerates from the bundle.

D Ancillary material

The repository contains the group libraries with their self-tests (`core/su2.py`, `core/su3.py`, `core/spin4.py`, `core/spin5.py`), the training drivers (`experiments/*_train.py`), the intervention and mechanism analyses (`experiments/*_causal.py`, `experiments/su2_mechanism.py`, `experiments/twosided.py`, `experiments/su2_fusion.py`, `experiments/hk_analysis.py`), and the results bundle (`summary/results_bundle.json`). The paper stands without these files: every claim above is proved in Appendix A, established by the reported measurements, or cited.